

12.4 Simultaneous bands for the regression line and surface

A fitted line is usually drawn inside a shaded region. If the region is made of pointwise intervals, each vertical slice covers the true mean with probability $1 - \alpha$, but the region does not cover the whole true line with that probability. A reader who asks “which lines are consistent with the data?” needs a simultaneous band, and for a linear model the Cauchy–Schwarz inequality gives one with exact coverage.

12.4.1 Pointwise is not simultaneous

Definition 12.4.1 — Simultaneous confidence band

Let $\mathcal{S} \subseteq \mathcal{C}(\mathbf{X}^\top)$ be a set of regressor vectors. Random functions $\ell(\mathbf{x}) = \ell(\mathbf{x}; \mathbf{Y})$ and $u(\mathbf{x}) = u(\mathbf{x}; \mathbf{Y})$ form a $1 - \alpha$ **simultaneous confidence band** for the regression function over $\mathcal{S}$ if

$$\Pr_{\beta, \sigma^2} \{ \ell(\mathbf{x}) \leq \mathbf{x}^\top \beta \leq u(\mathbf{x}) \text{ for every } \mathbf{x} \in \mathcal{S} \} \geq 1 - \alpha$$

for every (β, σ^2) .

Pointwise t intervals fail this: for the stack loss line of [Example \(A band for the stack loss line\)](#), the pointwise 95% intervals cover the true line over the observed range in only 0.8705 of simulated samples.

12.4.2 The Working–Hotelling band

Theorem 12.4.1 — Working–Hotelling band

Let $\mathcal{L}$ be a linear subspace of $\mathcal{C}(\mathbf{X}^\top)$ of dimension $k \geq 1$, and let $w_k = \sqrt{k F_\alpha(k, n - r)}$. Then

$$\Pr \{ |\mathbf{x}^\top \hat{\beta} - \mathbf{x}^\top \beta| \leq w_k s \sqrt{\mathbf{x}^\top \mathbf{G} \mathbf{x}} \text{ for all } \mathbf{x} \in \mathcal{L} \} = 1 - \alpha. \quad (12.4.1)$$

In particular:

- (a) with $\mathcal{L} = \mathcal{C}(\mathbf{X}^\top)$ and $k = r$, the functions $\mathbf{x}^\top \hat{\beta} \pm w_r s \sqrt{\mathbf{x}^\top \mathbf{G} \mathbf{x}}$ form an exact $1 - \alpha$ simultaneous band for the regression surface over all estimable $\mathbf{x}$;
- (b) for a straight line $E Y_i = \beta_0 + \beta_1 x_i$ with the x_i not all equal, the band

$$\hat{\beta}_0 + \hat{\beta}_1 x \pm \sqrt{2 F_\alpha(2, n - 2)} s \sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}}}, \quad x \in \mathbb{R},$$

contains the true line $\beta_0 + \beta_1 x$ for every real x with probability exactly $1 - \alpha$.

Proof. Step 1: from $\mathcal{L}$ to a subspace of $\mathcal{C}(\mathbf{X})$. The map $\mathbf{u} \mapsto \mathbf{X}^\top \mathbf{u}$ is one to one on $\mathcal{C}(\mathbf{X})$, since $\mathbf{X}^\top \mathbf{u} = \mathbf{0}$ with $\mathbf{u} \in \mathcal{C}(\mathbf{X})$ forces $\mathbf{u} \in \mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{X})^\perp = \{\mathbf{0}\}$. It maps $\mathcal{C}(\mathbf{X})$ onto $\mathcal{C}(\mathbf{X}^\top)$, because $\mathbf{X}^\top \boldsymbol{\rho} = \mathbf{X}^\top \mathbf{M} \boldsymbol{\rho}$. So $\mathcal{U} = \{\mathbf{u} \in \mathcal{C}(\mathbf{X}) : \mathbf{X}^\top \mathbf{u} \in \mathcal{L}\}$ is a subspace of $\mathcal{C}(\mathbf{X})$ of dimension k , and every $\mathbf{x} \in \mathcal{L}$ is $\mathbf{X}^\top \mathbf{u}$ for exactly one $\mathbf{u} \in \mathcal{U}$. For such a pair,

$$\begin{aligned} \mathbf{x}^\top \hat{\boldsymbol{\beta}} - \mathbf{x}^\top \boldsymbol{\beta} &= \mathbf{u}^\top (\mathbf{X} \hat{\boldsymbol{\beta}} - \mathbf{X} \boldsymbol{\beta}) = \mathbf{u}^\top \mathbf{M} (\mathbf{Y} - \mathbf{X} \boldsymbol{\beta}) = \mathbf{u}^\top \mathbf{v}, \\ \mathbf{x}^\top \mathbf{G} \mathbf{x} &= \mathbf{u}^\top \mathbf{X} \mathbf{G} \mathbf{X}^\top \mathbf{u} = \mathbf{u}^\top \mathbf{M} \mathbf{u} = \|\mathbf{u}\|^2, \end{aligned}$$

where $\mathbf{v} = \mathbf{P}_{\mathcal{U}}(\mathbf{Y} - \mathbf{X} \boldsymbol{\beta})$. The middle equality uses $\mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{M} \mathbf{Y}$ and $\mathbf{M} \mathbf{X} = \mathbf{X}$. The last step of the first chain uses $\mathbf{u} = \mathbf{M} \mathbf{u} = \mathbf{P}_{\mathcal{U}} \mathbf{u}$, and the second chain uses Theorem 6.3.7.

Step 2: the supremum. By the Cauchy–Schwarz inequality, for $\mathbf{u} \in \mathcal{U}$, $\mathbf{u} \neq \mathbf{0}$,

$$\frac{(\mathbf{x}^\top \hat{\boldsymbol{\beta}} - \mathbf{x}^\top \boldsymbol{\beta})^2}{\mathbf{x}^\top \mathbf{G} \mathbf{x}} = \frac{(\mathbf{u}^\top \mathbf{v})^2}{\|\mathbf{u}\|^2} \leq \|\mathbf{v}\|^2,$$

with equality at $\mathbf{u} = \mathbf{v}$ when $\mathbf{v} \neq \mathbf{0}$, and $\mathbf{v} \in \mathcal{U}$. So the supremum of the left side over $\mathbf{x} \in \mathcal{L} \setminus \{\mathbf{0}\}$ equals $\|\mathbf{v}\|^2$ and is attained.

Step 3: its distribution. $\mathbf{Y} - \mathbf{X} \boldsymbol{\beta} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I})$, and $\mathbf{P}_{\mathcal{U}}$ is symmetric idempotent of rank k , so $\|\mathbf{v}\|^2 / \sigma^2 \sim \chi^2(k)$ (Theorem 4.3.2(a)). Since $\mathcal{U} \subseteq \mathcal{C}(\mathbf{X})$, $\mathbf{P}_{\mathcal{U}} = \mathbf{P}_{\mathcal{U}} \mathbf{M}$, so $\mathbf{v}$ is a function of $\mathbf{M} \mathbf{Y}$ and is independent of SSE (Theorem 7.5.1(c)). Hence $\|\mathbf{v}\|^2 / (k \sigma^2) \sim F(k, n - r)$ (Definition 4.1.2), and the event in Eq. 12.4.1, which is $\|\mathbf{v}\|^2 \leq w_k^2 s^2$ by Step 2, has probability $1 - \alpha$. (At $\mathbf{x} = \mathbf{0}$ both sides of the inequality are zero.)

Part (a) is the case $\mathcal{L} = \mathcal{C}(\mathbf{X}^\top)$, $\mathcal{U} = \mathcal{C}(\mathbf{X})$.

Part (b). Here $r = k = 2$ and $\mathcal{L} = \mathbb{R}^2$, but the band uses only the vectors $\mathbf{x} = (1, x)^\top$. The ratio in Step 2 is unchanged when $\mathbf{x}$ is multiplied by a nonzero scalar, so its supremum over $\{(1, x)^\top : x \in \mathbb{R}\}$ equals its supremum over all $\mathbf{x} = (a, b)^\top$ with $a \neq 0$. By Step 2 the supremum over all of $\mathbb{R}^2$ is attained at $\mathbf{x}^* = \mathbf{X}^\top \mathbf{v}$, whose first coordinate is $\mathbf{1}^\top \mathbf{v}$. The random variable $\mathbf{1}^\top \mathbf{v}$ is normal with variance $\sigma^2 \|\mathbf{P}_{\mathcal{U}} \mathbf{1}\|^2 = \sigma^2 n > 0$, because $\mathbf{1} \in \mathcal{C}(\mathbf{X}) = \mathcal{U}$. So with probability one $\mathbf{x}^*$ has a nonzero first coordinate, the two suprema are equal, and the band event has the probability computed in Step 3. The formula for $\mathbf{x}^\top \mathbf{G} \mathbf{x}$ is Eq. 5.5.2 divided by σ^2 .

■

The proof has one idea: every linear function of the fitted mean is an inner product with $\mathbf{v}$, and the worst case over all directions is $\|\mathbf{v}\|$. With $\mathcal{L} = \mathcal{C}(\mathbf{X}^\top)$ the band holds exactly when the confidence ball of Proposition 12.2.2 contains $\mathbf{X} \boldsymbol{\beta}$: the band is that ball, viewed one linear functional at a time. Applied to general families of estimable functions, the argument gives Scheffé's intervals, which Chapter 13 develops (Theorem 13.2.2) and compares with Bonferroni and Tukey.

The straight-line band is hyperbolic, narrowest at $\bar{x}$. Although x varies along a line, the multiplier uses $k = 2$, because the band must hold for intercept and slope together.

12.4.3 How much wider?

The band is wider than the pointwise intervals by the factor $w_k/t_{n-r,\alpha/2}$, the same at every $\mathbf{x}$. The multipliers $w_k = \sqrt{kF_{0.05}(k, \nu)}$ for 95% coverage are:

k	$\nu = 5$	$\nu = 10$	$\nu = 20$	$\nu = 60$	$\nu = \infty$
1	2.571	2.228	2.086	2.000	1.960
2	3.402	2.865	2.643	2.510	2.448
3	4.028	3.335	3.049	2.876	2.795
5	5.025	4.078	3.682	3.441	3.327
10	6.881	5.457	4.845	4.464	4.279

The row $k = 1$ is the t multiplier; the column $\nu = \infty$ is $\sqrt{\chi_{0.05}^2(k)}$, for known σ . For a straight line the band is about a quarter wider than the pointwise intervals, for ten coefficients more than twice as wide: for large k simultaneity costs roughly a factor $\sqrt{k}$.

Example. (A band for the stack loss line)

Regress stack loss on air flow for the 21 days of Brownlee's plant data (Example (Stack loss)), ignoring the other two regressors. The fitted line is $-44.132 + 1.0203x$ with $s = 4.098$, $\bar{x} = 60.43$ and $S_{xx} = 1681.1$. The pointwise multiplier is $t_{19,0.025} = 2.093$, and the Working–Hotelling multiplier is $\sqrt{2F_{0.05}(2, 19)} = 2.654$, larger by the factor 1.268. Figure 12.4.1 shows the data, the band, the pointwise intervals, and the 95% prediction intervals of Section 12.3.

Near the centre, at $x = 60$, the fitted mean is 17.09, and the half-widths are 1.87 (pointwise), 2.38 (band) and 8.78 (prediction). At $x = 80$, the edge of the data, the fitted mean is 37.49 and the half-widths are 4.50, 5.71 and 9.69.

In 40000 data sets simulated from the fitted line, the band contained the true line over the whole real line in a proportion 0.9500 of the samples and over the observed range $50 \leq x \leq 80$ in 0.9537. The pointwise intervals covered at $x = 60$ in 0.9517 of the samples, but covered the line over the whole observed range in only 0.8705.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.stackloss.load_pandas().data
x = data["AIRFLOW"].to_numpy()
y = data["STACKLOSS"].to_numpy()
n = len(y)
X = np.column_stack([np.ones(n), x])
C = np.linalg.inv(X.T @ X)
beta_hat = C @ X.T @ y
s = np.sqrt(np.sum((y - X @ beta_hat) ** 2) / (n - 2))

grid = np.linspace(45, 85, 401)
Z = np.column_stack([np.ones_like(grid), grid])
fit = Z @ beta_hat
se_mean = s * np.sqrt(np.einsum("ij,jk,ik->i", Z, C, Z)) # s * sqrt(1/n + (x -
xbar)**2 / Sxx)
```

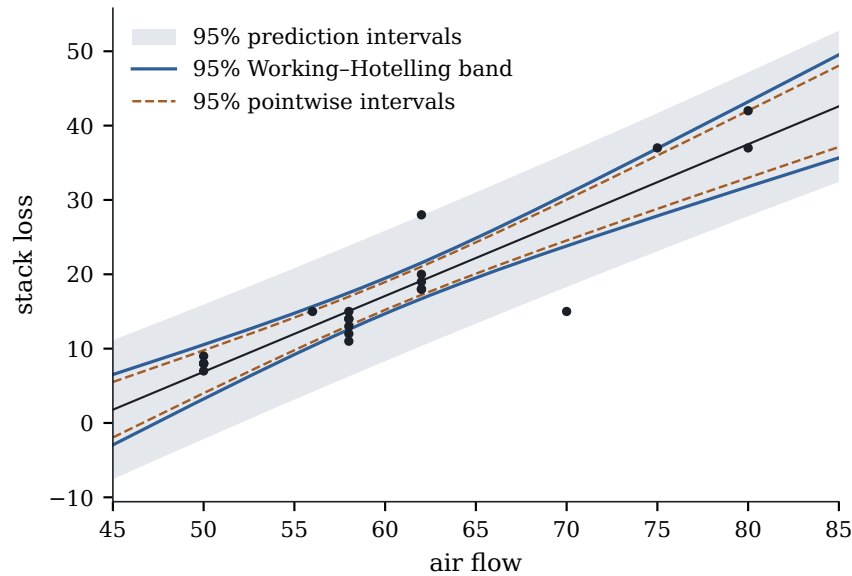


Figure 12.4.1: Stack loss against air flow: least squares line, 95% Working–Hotelling band (solid), pointwise 95% intervals for the mean (dashed) and pointwise 95% prediction intervals (shaded).

```
t_pt = stats.t.ppf(0.975, n - 2)                                # pointwise multiplier
w_wh = np.sqrt(2 * stats.f.ppf(0.95, 2, n - 2))                # Working-Hotelling
multiplier
pointwise = (fit - t_pt * se_mean, fit + t_pt * se_mean)
band = (fit - w_wh * se_mean, fit + w_wh * se_mean)
se_new = np.sqrt(s**2 + se_mean**2)
prediction = (fit - t_pt * se_new, fit + t_pt * se_new)
print(f"slope {beta_hat[1]:.4f}, s = {s:.3f}; multipliers: t {t_pt:.3f}, Working-
      Hotelling {w_wh:.3f}")
```

[Run this code online](#)

12.4.4 Bands over part of the space, and other shapes

The band holds over the whole line. If only $x \in [a, b]$ matters it is conservative (the simulated 0.9537), and an exact band over $[a, b]$ uses a smaller multiplier (Wynn and Bloomfield 1971). The hyperbolic shape is also a choice: bands of constant width, or with straight edges, come from replacing the Euclidean norm in Step 2 by another norm (Gafarian 1964; Liu 2010). And if only some regressors vary, $\mathcal{L}$ is smaller, and so is k .

12.4.5 Exercises

12.4.5.1 A. Check your understanding

Exercise A1.

For a straight line with many residual degrees of freedom, compute the ratio of the

Working–Hotelling multiplier to the pointwise one at $\alpha = 0.05$. How does the ratio change at $\alpha = 0.01$?

Solution. As $\nu \rightarrow \infty$, $2F_\alpha(2, \nu) \rightarrow \chi_\alpha^2(2) = -2 \log \alpha$, since $\chi^2(2)$ is exponential with mean 2. At $\alpha = 0.05$ the ratio is $\sqrt{2 \log 20} / z_{0.025} = 2.448 / 1.960 = 1.249$. At $\alpha = 0.01$ it is $\sqrt{2 \log 100} / z_{0.005} = 3.035 / 2.576 = 1.178$. The relative cost of simultaneity falls as the level becomes more demanding. As $\alpha \rightarrow 0$, both squared multipliers, $z_{\alpha/2}^2$ and $\chi_\alpha^2(2) = 2 \log(1/\alpha)$, grow like $2 \log(1/\alpha) + O(\log \log(1/\alpha))$, so the ratio of the multipliers tends to 1.

Exercise A2.

For the line through the origin $EY_i = \beta x_i$, show that a simultaneous band for βx over all real x is $\hat{\beta}x \pm t_{n-1, \alpha/2} s|x| / \sqrt{\sum x_i^2}$, with no widening at all. Explain this with the dimension k of Theorem 12.4.1.

Solution. Here $\mathcal{C}(\mathbf{X}^\top) = \mathbb{R}$, so $k = 1$ and $w_1 = t_{n-1, \alpha/2}$. Directly: the band holds at every x iff it holds at $x = 1$, since both sides scale with $|x|$.

12.4.5.2 B. Practice

Exercise B1.

For the straight line, show that the Working–Hotelling band contains the line $b_0 + b_1x$ for every x iff (b_0, b_1) lies in the $1 - \alpha$ confidence ellipse for (β_0, β_1) of Theorem 12.2.1. Deduce that the band is the envelope of all lines whose coefficients lie in the ellipse.

Solution. Put $\mathbf{u} = (\hat{\beta}_0 - b_0, \hat{\beta}_1 - b_1)^\top$ and $\mathbf{W} = (\mathbf{X}^\top \mathbf{X})^{-1}$. The line $b_0 + b_1x$ is inside the band at x iff $|(1, x)\mathbf{u}| \leq w_2 s \sqrt{(1, x)\mathbf{W}(1, x)^\top}$. By scale invariance this holds for all x iff it holds for all $\mathbf{a} = (a_1, a_2)^\top$ with $a_1 \neq 0$, and by continuity iff it holds for all $\mathbf{a}$. By Lemma 12.2.3 that is $\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u} \leq w_2^2 s^2$, which is the ellipse. At each x , the upper edge of the band is the maximum of $b_0 + b_1x$ over the ellipse (Corollary 12.2.4), so the band is the envelope.

Exercise B2.

For quadratic regression $EY_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2$, the natural band over all x uses $\mathbf{x} = (1, x, x^2)^\top$, a curve in $\mathbb{R}^3$ whose directions do not fill $\mathbb{R}^3$. Show that the band with multiplier w_3 has coverage at least $1 - \alpha$ over all x , and argue that its coverage is strictly larger than $1 - \alpha$. (Hint: when is the maximizing direction $\mathbf{X}^\top \mathbf{v}$ of Step 2 a multiple of some $(1, x, x^2)^\top$?)

Solution. The band event over all $\mathbf{x} \in \mathbb{R}^3$ has probability $1 - \alpha$ and is contained in the event over the curve, so the coverage is at least $1 - \alpha$. Write $S = \|\mathbf{v}\|^2$ and S_c for the supremum of the ratio of Step 2 over the curve. The ratio depends on $\mathbf{v}$ only through S and the direction $\mathbf{v}/\|\mathbf{v}\|$, so $S_c = \rho S$, where $\rho \leq 1$ is a continuous function of the direction alone. Equality $\rho = 1$ requires either that $\mathbf{X}^\top \mathbf{v}$ be a multiple of some $(1, x, x^2)^\top$, that is, its coordinates (c_0, c_1, c_2) satisfy $c_0 c_2 = c_1^2$, or that $\mathbf{X}^\top \mathbf{v}$ be a multiple

of $(0, 0, 1)^\top$, the limit of $(1, x, x^2)^\top/x^2$ as $x \rightarrow \infty$, where the supremum is approached but not attained. Both are events of probability zero. For a spherical normal vector in $\mathcal{U}$, the length and the direction are independent, and both are independent of s^2 . So the coverage over the curve exceeds $1 - \alpha$ by $\Pr\{w_3^2 s^2 < S \leq w_3^2 s^2 / \rho\}$, which is positive because S/σ^2 has a positive density on $(0, \infty)$ and $\rho < 1$ almost surely.

12.4.5.3 C. Going deeper

Exercise C1.

A single new observation $Y_0 = \mathbf{x}_0^\top \boldsymbol{\beta} + \varepsilon_0$ will be taken at a regressor vector $\mathbf{x}_0 \in \mathcal{C}(\mathbf{X}^\top)$ that is not known in advance. Show that the band $\mathbf{x}^\top \hat{\boldsymbol{\beta}} \pm \sqrt{(r+1)F_\alpha(r+1, n-r)} s \sqrt{1 + \mathbf{x}^\top \mathbf{G}\mathbf{x}}$, $\mathbf{x} \in \mathcal{C}(\mathbf{X}^\top)$, contains Y_0 with probability at least $1 - \alpha$, whatever $\mathbf{x}_0$ turns out to be. (Hint: apply Step 2 of the proof to the $(r+1)$ -dimensional vector $(\mathbf{v}, \varepsilon_0)$.)

Solution. For $\mathbf{x} = \mathbf{X}^\top \mathbf{u}$ with $\mathbf{u} \in \mathcal{C}(\mathbf{X})$, $Y_0 - \mathbf{x}^\top \hat{\boldsymbol{\beta}} = \varepsilon_0 - \mathbf{u}^\top \mathbf{v}$, with $\mathbf{v} = \mathbf{M}(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$, and $1 + \mathbf{x}^\top \mathbf{G}\mathbf{x} = 1 + \|\mathbf{u}\|^2$. By Cauchy–Schwarz in $\mathcal{C}(\mathbf{X}) \times \mathbb{R}$, $(\varepsilon_0 - \mathbf{u}^\top \mathbf{v})^2 \leq (1 + \|\mathbf{u}\|^2)(\varepsilon_0^2 + \|\mathbf{v}\|^2)$. The variable $(\varepsilon_0^2 + \|\mathbf{v}\|^2)/\sigma^2$ is $\chi^2(r+1)$, independent of SSE, so $(\varepsilon_0^2 + \|\mathbf{v}\|^2)/((r+1)s^2) \sim F(r+1, n-r)$, and on the event that it is at most $F_\alpha(r+1, n-r)$, which has probability $1 - \alpha$, the band contains Y_0 at every $\mathbf{x}$, in particular at $\mathbf{x}_0$.

Exercise C2.

Two straight lines are fitted in one model, with separate intercepts and slopes and a common σ^2 . Construct simultaneous bands for both lines with joint coverage at least $1 - \alpha$ in two ways: (i) [Theorem 12.4.1](#) with $\mathcal{L} = \mathbb{R}^4$, (ii) a band for each line with $\mathcal{L} = \mathbb{R}^2$ at level $\alpha/2$, combined by the Bonferroni inequality. Compare the multipliers for $n - r = 20$ and explain which is better and why.